

Phase on the Merge: Generated, and Its Cost

Two steps coupling proto-phase to the additive merge

Alexander Pisani & Claude (Anthropic) · June 2026 · companion note

Framing. Continuing the erasure-only cartography: map values, note where they line up, assert no mechanism of the genesis system. $\text{ln}(\cdot)$ is the generic Landauer signature, so the content is always the multiplicity inside it. Claims graded Fact / Reading / Lead. Verified by `phase_from_merge.py`, `phase_cost.py`. Builds directly on *The Additive Merge: Heat, Storage, and the Reverse* (the reverse-scattering and the Weyl shadow) and *Phase from the Probe* ($\sqrt{\text{NOT}}$ from succession and the probe).

1. The phase is generated, not imposed

The reverse-scattering of a merge to $\mathbf{n}$ lives on the $(\mathbf{n}+1)$ -dimensional preimage space and carries the Weyl algebra, but the scattering itself is phase-blind — probabilities, no $\mathbf{i}$. Rather than paint phases on by hand, we asked whether the two genesis operations already in hand generate the phase on this space:

- $\mathbf{S}$ = succession-shift on the split index, $|\mathbf{k}\rangle \rightarrow |\mathbf{k}+1 \bmod (\mathbf{n}+1)\rangle$ (the Weyl $\mathbf{X}$);
- $\mathbf{D}$ = origin meet-atom probe, $|0\rangle\langle 0|$.

They fail to commute, and the algebra they generate contains the Weyl clock $\mathbf{Z}$ (the phase) for every $\mathbf{n} = 1..5$, since the diagonal projectors $|\mathbf{k}\rangle\langle \mathbf{k}| = \mathbf{S} \mathbf{D} \mathbf{S}$ span all diagonals including $\mathbf{Z} = \sum |\mathbf{k}\rangle\langle \mathbf{k}|$. At the genesis bit the commutator is exactly the phase generator:

$$[\mathbf{S}, \mathbf{D}] = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i \sigma_y,$$

the $\mathbf{i}$ of $\sqrt{\text{NOT}}$. So the phase the scattering lacked is produced by succession and the probe acting on the scattering's own space — the *Phase from the Probe* result re-situated where the merge needs it, and now general in $\mathbf{n}$, not only at $\mathbf{p}=2$. (*Non-commutation, $\mathbf{Z}$ generated from $\{\mathbf{S}, \mathbf{D}\}$, and $[\mathbf{S}, \mathbf{D}] = -i_y$ at the genesis bit: Fact.*)

This establishes only that the phase is **structurally available** — the generating operators are present. It does not show the merge *dynamics* produce a particular phased state; that remains open.

2. The phase cost is the merge-erasure

Put a definite phase on the scattering — a coherent superposition with the clock phases — and read the cost via decoherence:

n	classical mixture S	phased pure S	erase phase (decohere)	merge-erasure $\ln(n+1)$
1	0.6931	0	0.6931	0.6931
2	1.0986	0	1.0986	1.0986
4	1.6094	0	1.6094	1.6094
8	2.1972	0	2.1972	2.1972

The classical scattering is the maximally mixed state (entropy $\ln(n+1)$); a coherent phased superposition is pure (entropy 0); decohering it — losing the phase — creates exactly $\ln(n+1)$, matching the merge-erasure at every n . So **phase is not a separate cost stacked on the merge; it is the same $\ln(n+1)$, seen as the coherence whose loss is the merge’s irreversibility**. The additive merge is irreversible precisely because it decoheres; a phase-preserving merge would be reversible (zero erasure), with the split recoverable from the phase. This is the merge-erasure and the phase as one quantity, two faces. (*The entropies: Fact. “Merge irreversibility = phase loss; phase carries the erased split”: well-grounded Reading, standard decoherence-as-erasure.*)

3. Split-blindness: broken in the angle, not the cost — Goldbach still out of reach

Does phase let the ledger tell one split of 8 from another? Two parts, opposite answers:

- The per-split phase **angle** varies with the index k (40° steps at $n=8$): (1,7) at 40° , (3,5) at 120° . Splits are now distinguishable *by phase*.
- The per-split phase **cost** is $\ln 9$ for *every* split — uniform, split-blind, exactly like the merge-erasure. And the angle is *positional* (linear in k), not arithmetic: nothing marks (3,5) as a prime pair rather than “the split at index 3.”

So the cost remains split-blind, and the only per-split structure is a positional label. For a Goldbach-type question to be askable, prime pairs would have to be cost-extremal or phase-special; they are neither, because distinguishing them needs multiplicative structure (primality), which a purely additive proto-integer system does not contain. The system is incomplete in exactly the anticipated way. (*Per-split angle versus uniform cost: Fact. Goldbach out of reach pending arithmetic phase: honest assessment.*)

4. Grading and parked leads

- **Fact** — $[S,D] = 0$ and the Weyl phase Z generated from $\{S, D\}$ on the preimage space; $[S,D] = -i_y$ at the genesis bit; phase coherence holds $\ln(n+1)$, equal to the merge-erasure (mixture entropy $\ln(n+1)$, pure entropy 0, decoherence creates $\ln(n+1)$); per-split phase angle positional, per-split phase cost uniform.
- **Reading** — the merge’s irreversibility *is* decoherence, with phase the carrier of the erased split (the merge-erasure and the phase as one quantity, two faces); the $n=1$ commutator being “the same genesis bit” as the Pauli/ $\sqrt{\text{NOT}}$.
- **Lead** — that the merge *dynamics* (not just its algebra) produce a definite phased state; that interference fires once they do; coupling the additive merge to multiplicative structure so that phase could become arithmetic. These are the doors, none opened here.
- **Parked (Alex’s, deferred)** — whether the two Goldbach primes summing to an even number come out cost-extremal once arithmetic phase exists; *cheaper-to-generate more*

probable to observe. Both explicitly later.

5. Honesty boundary

Classical content is the Weyl/Pauli clock-shift algebra, Landauer (erasure = log multiplicity), and the von Neumann entropy of decoherence — all flagged. The contribution is showing that the merge’s reverse-scattering generates its own missing phase from succession and the probe, and that the cost of that phase equals the merge-erasure, so the merge’s irreversibility is its decoherence — while the cost stays split-blind, leaving the arithmetic (Goldbach) question correctly unreachable at this stage. No arithmetic question is claimed resolved and the genesis system’s mechanism is not asserted.

Originating direction (couple proto-phase to the merge; the phase-cost and Goldbach questions, both correctly parked) is Alexander Pisani’s; the generation and cost tests and this compilation were developed jointly with Claude (Anthropic). June 2026.